

Analogue Control

Engineering Technology

May, 2026

Analogue Control (A11845)

Short Title:	Analogue Control	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Automotive, Automation and IoT	
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Credits	5	Level: Advanced

Description of Module / Aims

The course aims to demonstrate the techniques used to analyse analogue control systems and to design controllers. These techniques will be demonstrated using examples of practical control systems and will be implemented by the student using MATLAB.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (WD_EAUTO_B)	4 / 7 / M
ANCO-0001	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	4 / 7 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 6 / M
ANCO-0001	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	3 / 5 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	3 / 6 / M
ANCO-0001	BEng (Hons) (WD_ETRIC_B)	4 / 7 / M
ANCO-0001	BSc (Hons) in Applied Electronics (WD_EELEC_B)	1 / 1 / M

Indicative Content

- System Modelling of selected Control systems using Classical Methods
- State Space Modelling
- Transient response analysis of Control systems
- Frequency Response Stability analysis
- Analogue Controller Design classical techniques
- Modern Controller design
- Three Term Controller

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Model analogue control systems.
2. Analyse analogue control systems in the time and frequency domains.
3. Design analogue controllers using classical design techniques.
4. Design analogue controllers using modern design techniques.
5. Use MATLAB for stability analysis and controller design.

Learning and Teaching Methods

- Lectures and Practicals
- - Lectures will involve delivery of theory and methodology with subsequent problem based learning
- - Practicals will involve use of MATLAB in analysis and design

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	
<i>Practical</i>	20%	5
<i>Lab Report</i>	10%	5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Supplementary Material

- Kuo, B.C. and F. Golnaraghi. *_Automatic Control Systems_*. 8th ed.. USA: Wiley, 2002.
- Nise, N.S. *_Control Systems Engineering_*. USA: Wiley, 2011.

Requested Resources

- Lecture Room: Loose Seated
- COMPUTER LAB: Networks Lab